



Technical Service Bulletin: TSB170078

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New Modular Common Rail System (MCRS) Fuel Pumps and External Pressure Relief Valve Service Kits for Gerotor Overpressure Caused by Fuel Filter Plugging: FC2261

New Modular Common Rail System (MCRS) Fuel Pumps and External Pressure Relief Valve Service Kits for Gerotor Overpressure Caused by Fuel Filter Plugging: FC2261

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK50 CM850 (All Versions)
- QSK50 CM2150 (All Versions)
- QSK60 CM2150 (All Versions)
- QSK60 CM850 (All Versions)

Issue

Symptom:

- Stumble, rough running, low power, and engine shut down
- Fault Code 2261 and 559
- Engine will **not** restart and check engine light illuminated
- High oil levels on lubricating oil dipstick

Root Cause:

- Malfunctions initiate at gerotor caused by gerotor outlet restriction. Example: Stage two fuel filter plugging

Verification

- Verify presence of Fault Code 2261 or associated fault codes and other evidence of stage two fuel filter increased restriction

Resolution

- A new service kit with the 11 bar [159.5 psi] external pressure relief valve in conjunction with the deactivation of the internal pressure relief valve of applicable fuel pumps is available to update engines in service.
- Industrial engines built at Daventry after 1 April 2016 and Marine engines built after 20 July 2016 are already equipped with the external pressure relief valve and new fuel pump with no internal pressure relief valve.

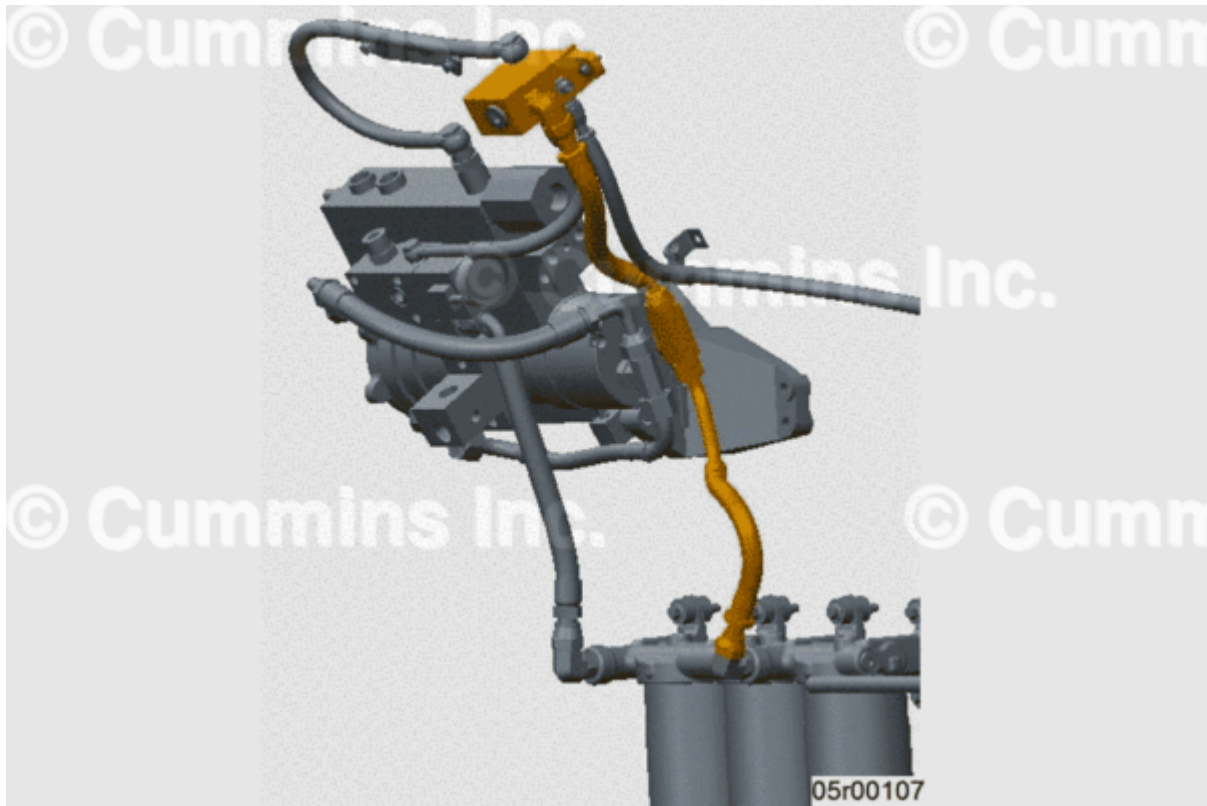


Figure 1, External Pressure Relief Valve Plumbing Example - QSK60 Industrial.



Figure 2, External Pressure Relief Valve Plumbing Example – QSK60 Marine.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
QSK50 External Pressure Relief Valve Service Kit	-	-	-	5405715
QSK60 External Pressure Relief Valve Service Kit	-	-	-	5405714
Pressure Relief Valve	-	-	-	4390272
Fuel Drain Manifold	4924663	No	No	4390206

Part Identification

The fuel pump will be marked after installing plug from kit, Part Number 5405714 or 5405715. The letter “P” will be stamped on outside of gerotor cover near plug location. Reference Figure 6 below. New fuel pumps listed in Table 1 can be distinguished from others by examining the gerotor housing and noting that the outboard internal pressure relief valve hex head plug is **not** present.

Service Instructions

External Pressure Relief Valve Upgrade Service Kit Installation Instructions:

Note : QSK60 engine kit, Part Number 5405714, does **not** contain elbows. Reuse gerotor outlet elbow; relocating elbow to the new fuel return block used in the kit.

- Verify all components are clean during installation.
- Remove fuel return block and gerotor outlet to stage two fuel filter head inlet plumbing.
- Install new pressure relief valve in gerotor outlet. Pressure relief valve **must** be installed such that normally open condition allows fuel to flow from gerotor outlet through external pressure relief valve and to fuel filters. Longer portion of pressure relief valve body oriented towards fuel return block. See Figure 3 below. **Torque Value:** 81 N•m [60 ft-lb]
- Install new hose assembly from pressure relief valve normally open outlet to fuel filter head inlet. Elbow hose fitting to fuel filter head **Torque Value:** 95 N•m [70 ft-lb]. Hose to external pressure relief valve **Torque Value:** 65 N•m [48 ft-lb].
- Install new hose assembly from relief valve normally closed outlet to new fuel return block. Elbow hose fitting in pressure relief valve **Torque Value:** 95 N•m [70 ft-lb]. Hose to union in fuel manifold **Torque Value:** 65 N•m [48 ft-lb].
- Reconnect remaining plumbing.
- Install internal pressure relief valve delete plug. See Internal Pressure Relief Valve Delete Plug Installation section below.

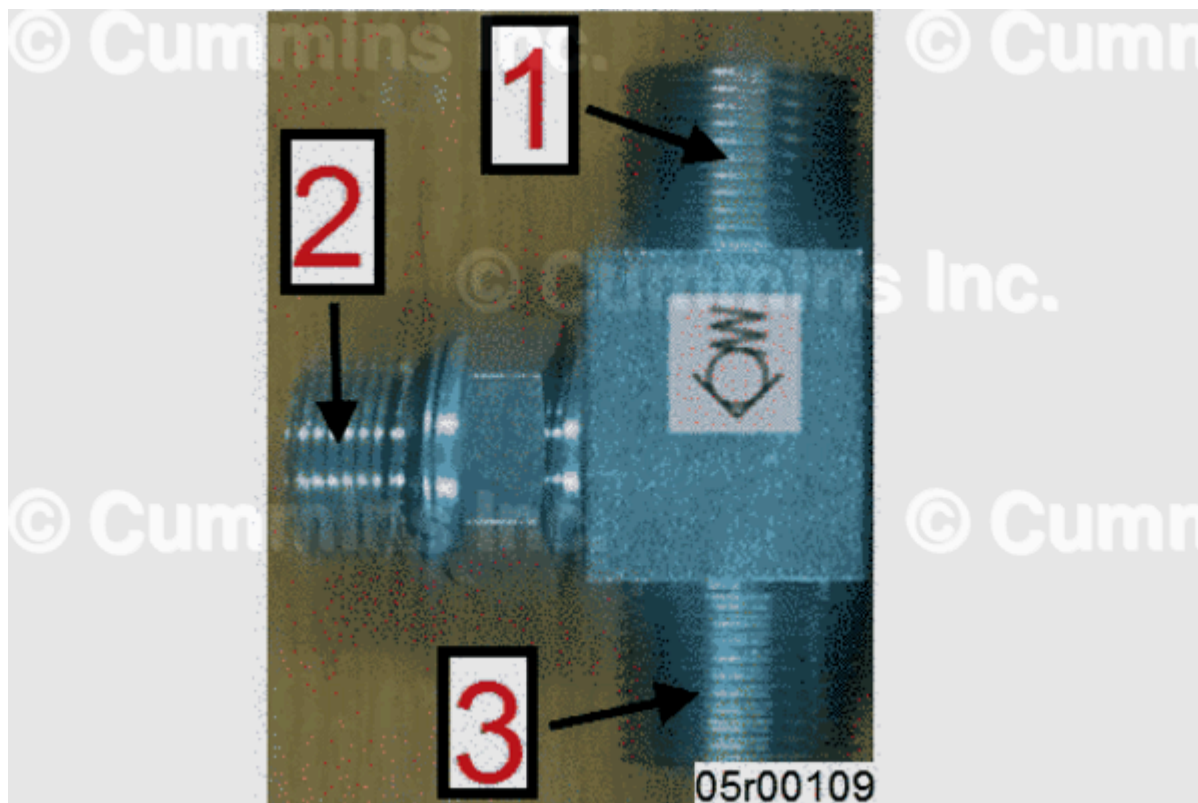


Figure 3, External Pressure Relief Valve.

1. To Fuel Return/Drain Block
2. Gerotor Outlet Port
3. To Stage 2 Fuel Filter Inlet

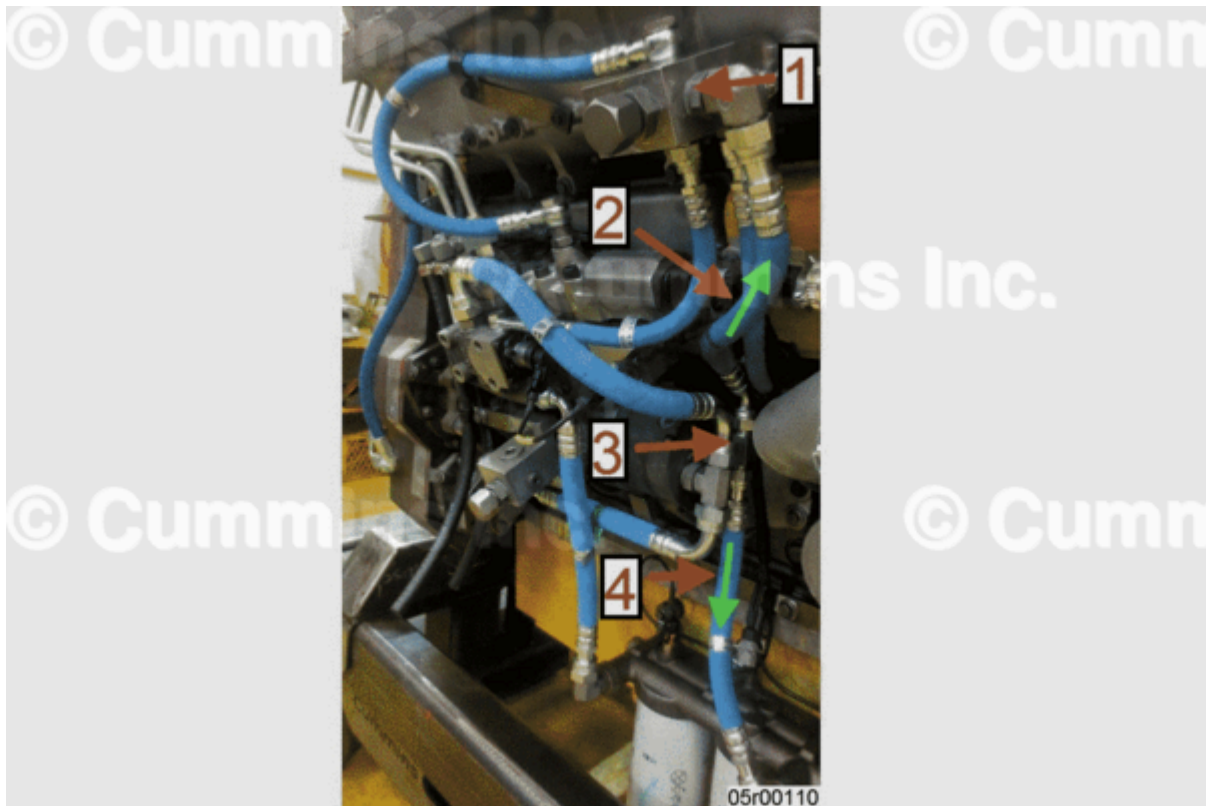


Figure 4, External Pressure Relief Valve Installation Layout.

1. External Pressure Relief Valve Check Valve towards Return Fuel Junction Block
2. Excess Gerotor Fuel Pressure
3. External Pressure Relief Valve
4. Gerotor Normal Flow Outlet to Stage 2 Fuel Filter Inlet

Internal Pressure Relief Valve Delete Plug Installation:

- Clean gerotor cover and top. Take note or mark insert depth of outboard facing hex head plug (internal pressure relief valve assembly) top.
- Remove outboard facing hex head plug from cover (internal pressure relief valve assembly) and remove copper sealing washer. Reuse copper sealing washer.
- Remove spring and valve piston.
- Verify bore is clear and clean. Do **not** allow debris to fall into bore.
- Lubricate provided interference plug with diesel fuel and install into lower bore with threaded portion pointed upwards. Verify plug is centered squarely in bore.
- Use hex head plug as an install tool. The copper washer is **not** to be installed at this time. Turn hex head plug into threaded bore until hex head plug bottoms out.
- Remove hex head plug, install copper washer, and install hex head plug into threaded bore. The head of hex head plug **must** be at same depth as noted earlier.

Torque Value: 54 n·m [40 ft-lb]



Figure 5, Internal Pressure Relief Valve Delete Plug Detail.



Figure 6, Example Fuel Pump Marking After Delete Plug Install.

Date	Details
2017-6-16	Module Created

Last Modified: 07-Jul-2017
